

Technology Stack

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, CSS3, Bootstrap	Used to create a responsive, user-friendly web interface for collecting user inputs and displaying the predicted HDI result.

Technology Stack Details

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Flask	Flask is used to develop the backend application, handle user requests, load the trained machine learning model, and generate HDI predictions.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV Dataset (HDI.csv), Pickle (.pkl)	The HDI dataset is stored as a CSV file for training, while the trained Linear Regression model is stored as a Pickle file for future predictions.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Render	Render is used to deploy the Flask application online, making it accessible through a web browser without requiring local execution.

5	Version Control & CI/CD	Git, GitHub	Git is used for version control, and GitHub is used to manage the source code, track changes, and collaborate on project development.
	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Pickle	Scikit-learn is used for machine learning model development, Pandas and NumPy for data processing, Matplotlib and Seaborn for data visualization, and Pickle for saving the trained model.
	<i>(e.g., Stripe, Twilio, Google Maps)</i>		